

InterviewAI

AI-Powered Interview Simulator — Business Model Canvas & Project Overview

Team: The Angels

Team Leader: Peter Maged

Track: AI & Intelligent Systems 2026

1. Executive Summary

Many job candidates never get enough realistic interview practice or personalized feedback before the interviews that decide their careers. InterviewAI is an AI-powered platform that simulates a real, adaptive job interview: it asks context-aware questions grounded in the candidate's resume and target job description, transcribes spoken or typed answers, evaluates them against criteria appropriate to the question type, analyzes speaking pace, filler words, and on-device body language, and returns a detailed, actionable improvement report — closer to a real interview and a real coach than a static question list or a generic chatbot.

The product is built on Next.js and deployed on Vercel, uses PostgreSQL (Neon) via Prisma for data, Clerk for authentication, and Google's Gemini API to power question generation, answer evaluation, and final reporting.

2. Problem

- Most candidates rehearse alone, with no realistic simulation of interview pressure or interaction.
 - Generic question lists don't adapt to what the candidate actually says, so practice doesn't build real interview reflexes.
 - Human mock interviews are valuable but expensive, hard to schedule, and not always available in the candidate's preferred language.
 - Feedback, when available at all, is often a single vague score with no explanation of what to actually change.
- ### 3. Solution
- An adaptive AI interviewer that asks one question at a time and generates realistic follow-ups based on the candidate's own answers.
 - Resume and job-description upload (PDF, DOCX, or text) so questions are grounded in the candidate's real experience and the specific role.
 - Voice or typed answers, live transcription, and speech-pace/filler-word analysis.
 - On-device, privacy-preserving body-language analysis (eye contact, posture, expression) — no camera frames ever leave the browser.
 - Per-answer AI evaluation with concrete strengths, weaknesses, suggestions, a STAR-method check, and a rewritten model answer.
 - A full performance report with a skill radar, an answer-by-answer breakdown, and a personalized practice plan.

4. Target Users & Market

Primary users: Students, fresh graduates, job seekers, developers, and professionals preparing for HR, behavioral, or technical interviews.

Target market: Job seekers in Egypt and the MENA region first, extending globally — especially candidates pursuing technology and software-related roles, and institutions (universities, bootcamps, career centers) that prepare them.

5. Business Model Canvas

The canvas below summarizes how InterviewAI creates, delivers, and captures value.

Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
● Google (Gemini API — core AI infrastructure) ● Neon (managed PostgreSQL hosting) ● Clerk (authentication & session management) ● Vercel (hosting & CI/CD) ● University career centers & coding bootcamps (distribution) ● Hackathon/developer communities (early adopters & feedback)	● Prompt engineering & continuous tuning of question generation and evaluation rubrics ● Product development (adaptive interview engine, body-language analysis, resume parsing) ● Content curation (question bank, per-question-type rubrics, bilingual content) ● Privacy & security maintenance (auth, data ownership, on-device processing)	● Realistic, adaptive AI interviews that react to what you actually say — not a static question list ● Actionable, structured feedback: per-criterion scores, strengths/weaknesses, STAR breakdown, and a rewritten model answer — not just a number ● Questions and follow-ups grounded in your real resume and the target job description ● Full bilingual support (English & Egyptian-ready Arabic) for an underserved MENA market ● Privacy-first: on-device body-language analysis, camera frames never uploaded, recording is opt-in ● Available 24/7 with no scheduling,	● Self-service product — sign up and start practicing independently ● Personalized progress tracking (history, trends, readiness score) builds a returning-user habit ● Community & hackathon feedback loop shaping the roadmap ● Future: light-touch coaching upsell and cohort features for institutions	● University students & fresh graduates preparing for their first interviews ● Job seekers changing careers, roles, or industries ● Software developers & engineers preparing for technical/system-design interviews ● Non-native English speakers who want to practice in Arabic first ● Career centers, bootcamps, and training providers (B2B distribution)

	User acquisition & community engagement	cost, or awkwardness of arranging a human mock interviewer	
	Key Resources - Proprietary prompt/rubric system for interview generation and evaluation - Full-stack codebase (Next.js, Prisma, PostgreSQL, Clerk, Gemini integration) - Google Gemini API access - Founding team's full-stack + applied-AI engineering expertise - Early user trust built on a privacy-first design		Channels - The web app itself (ai-interview-simulator-ten-navy.vercel.app) - LinkedIn and job-seeker social communities - University career centers and student clubs - Hackathon and developer-community exposure (e.g. Vibe Coding Arena) - Word of mouth and organic search
Cost Structure - AI inference costs (Gemini API calls, scale with active users and session length) - Cloud infrastructure: hosting (Vercel), database (Neon Postgres), authentication (Clerk) - Ongoing engineering & product development time			Revenue Streams Freemium: free core practice, paid tier for unlimited sessions, deeper analytics, and history

● Future: customer acquisition, support, and institutional-partnership costs	● B2B licensing: seat-based access for universities, bootcamps, and career centers ● Premium add-ons: downloadable certified performance reports, priority AI access ● Potential referral/placement partnerships with job boards and recruiters
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6. Product & Technology Snapshot

Frontend: Next.js 15 (App Router), React, TypeScript, Tailwind CSS

Authentication: Clerk — hosted sign-in, session-protected dashboard routes

Database: PostgreSQL hosted on Neon, accessed through Prisma ORM

AI: Google Gemini API — adaptive question generation, answer evaluation, and final report generation

Resume parsing: unpdf (PDF) and mammoth (DOCX), extracted server-side

Computer vision: @vladmandic/face-api (TensorFlow.js), running entirely client-side for body-language analysis

Hosting: Vercel

Live demo: <https://ai-interview-simulator-ten-navy.vercel.app/>

GitHub repository: <https://github.com/Mr-PeterMaged/AI-Interview-Simulator>

7. Team

Team name: The Angels

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